



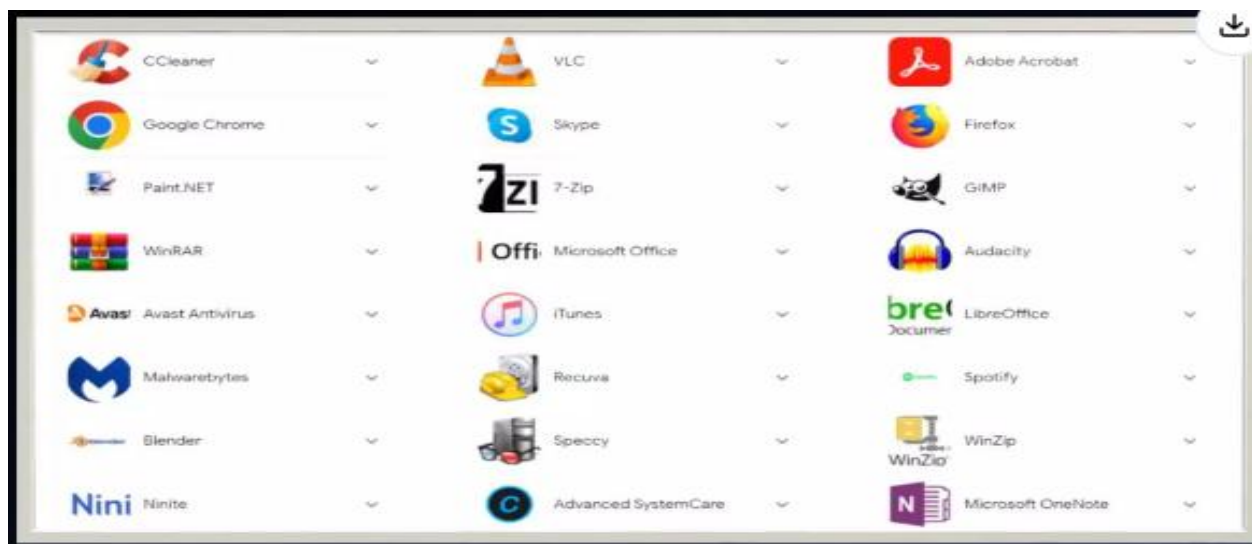
Application Development and Emerging Technologies

**Bachelor of Technical-Vocational Teacher Education
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What is Software

- Software is capable of performing many task, as opposed to hardware which can only perform mechanical task, that they are designed for. Software provides the means for accomplishing many different task with the same basic hardware.



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Classes of Software

- System Software
- Application Software
- Programming Software

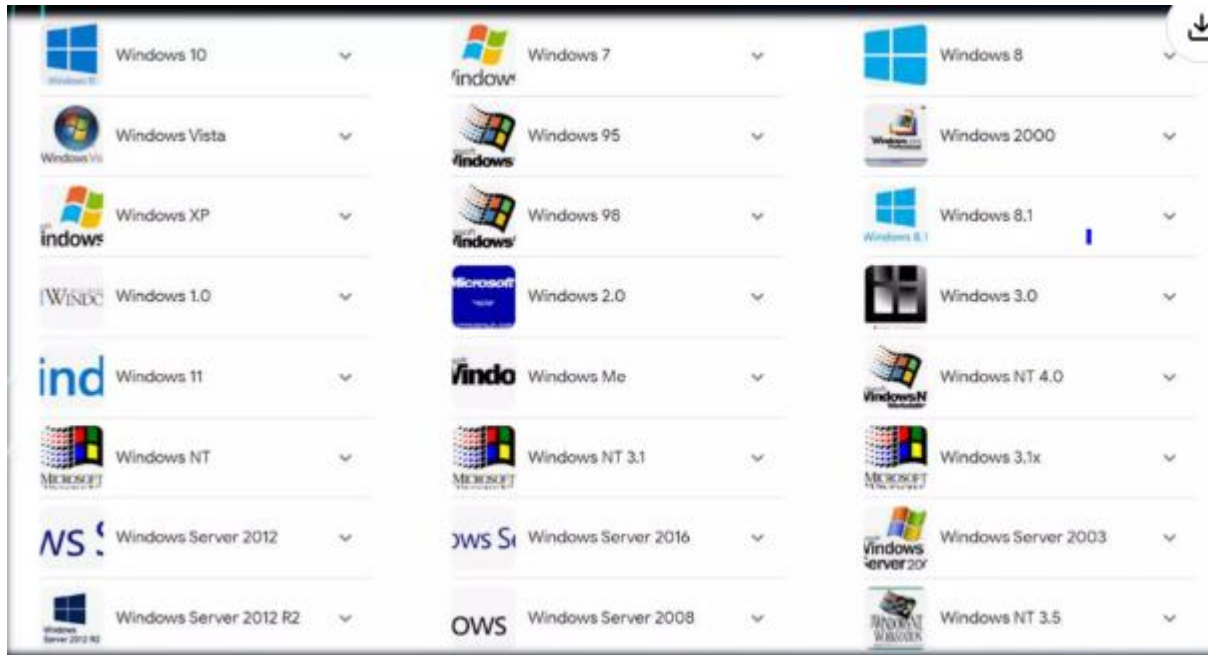


Classes of Software

- **System Software** – Helps run the computer hardware and computer system itself. System software includes operating system, device drivers, diagnostic tools and more. System software is almost always pre installed on your computer.



Classes of Software



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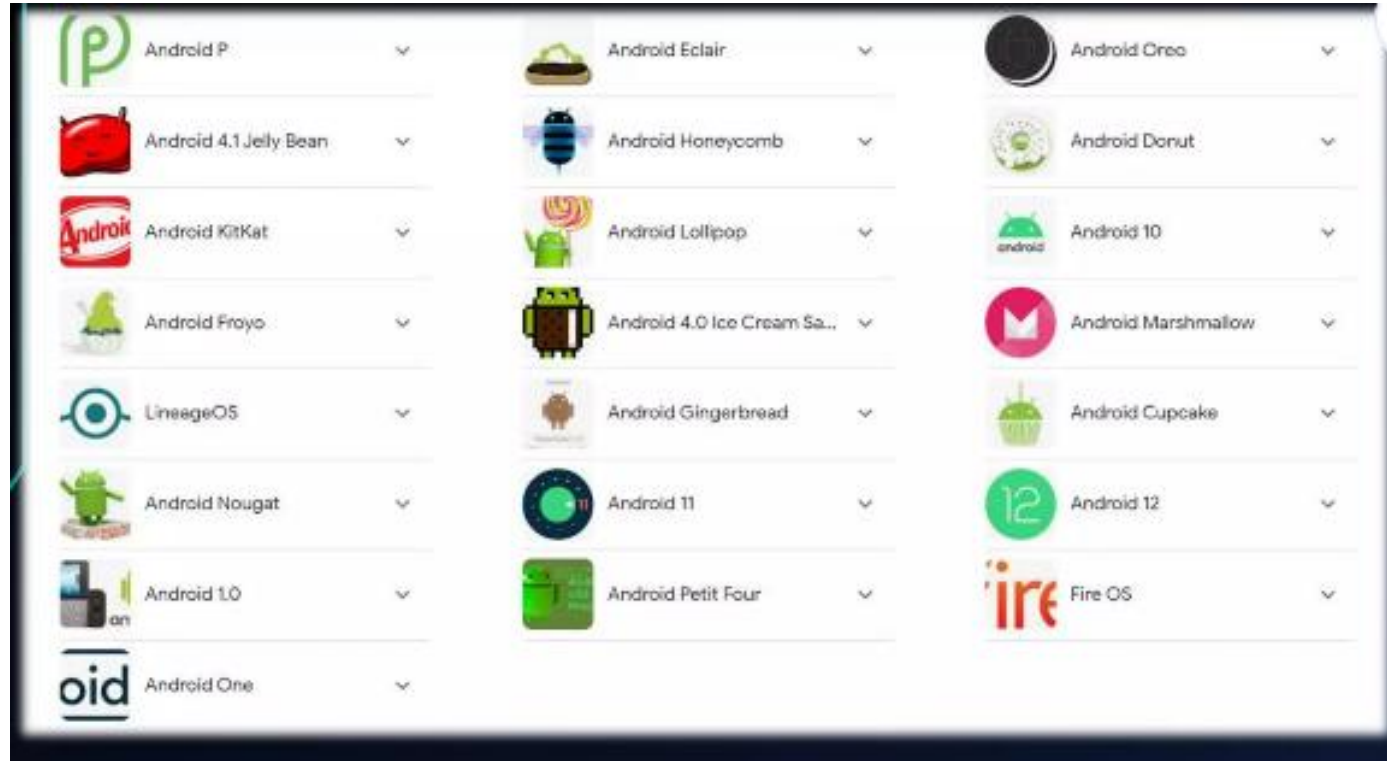
Classes of Software



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Classes of Software



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Classes of Software

- **Application Software**– Allow users to accomplish one or more task. It includes word processing, web browsing, and almost any other task for which might install software.



Classes of Software

General Purpose Application Software



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Classes of Software

- **Programming Software**– A set of tool to aid developers in writing program. The various tools available are compilers, linkers, debuggers, interpreters and text editors.



Classes of Software



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Basic Principles

- **Software**, commonly known as programs or apps, consist of all the instructions that tell the hardware how to perform task,
- These instructions come from a software developer in the form that will be accepted by the platform.
- For example, a program that is designed for windows operating system will only work for that specific OS.



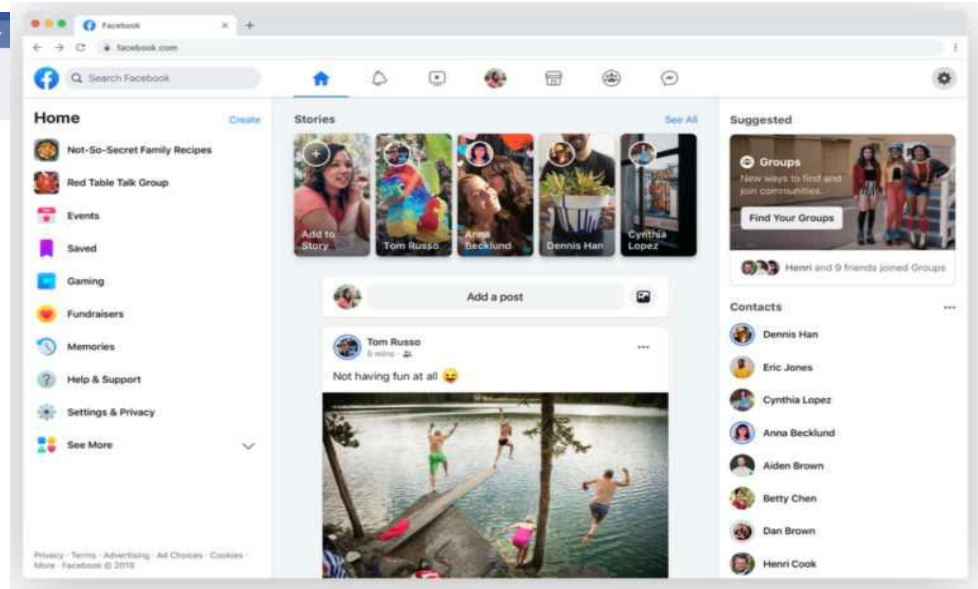
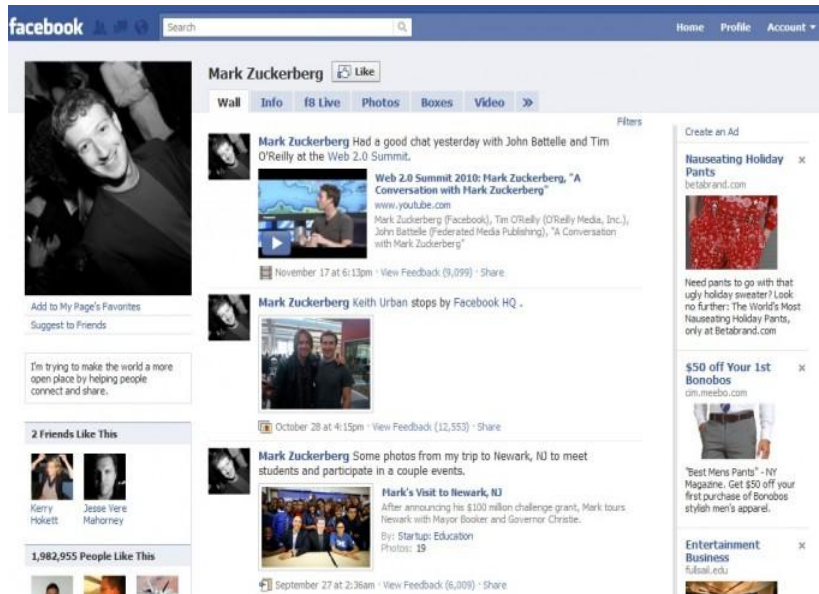


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Eight Laws of Software Evolution

- **1. Continuing Change** – A software system must continue to adapt to real world changes, else it becomes progressively less usefull.



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Eight Laws of Software Evolution

- **2. Increasing Complexity – A software system evolves, its complexity tends to increase unless work is done to maintain or reduce it.**

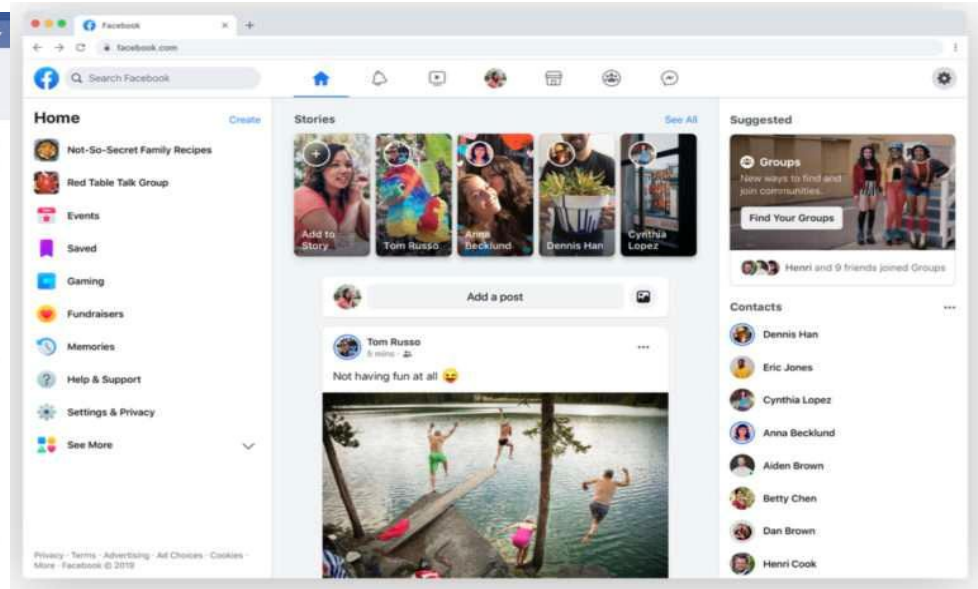
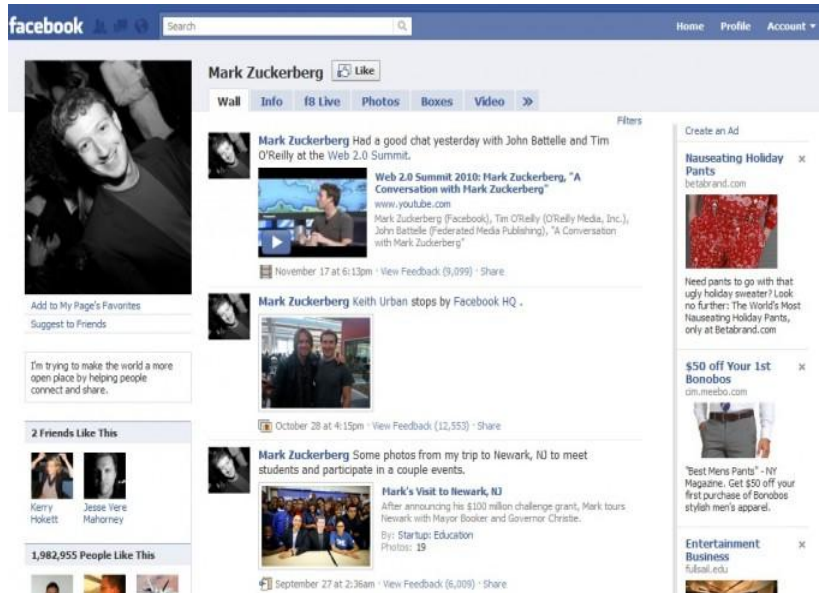


Eight Laws of Software Evolution

- **3. Conservation of Familiarity** – the familiarity with the software or the knowledge about how it was developed, why was it developed in that particular manner. Must be retain at any cost to implement the changes in the system



Eight Laws of Software Evolution



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Eight Laws of Software Evolution

- **4. Continuing Growth** – in order for a system intended to resolve, some business problem, its size of implementing the changes grows according to the lifestyle changes of the business.



Eight Laws of Software Evolution

- **5. Reducing Quality**– A software system declines in quality unless rigorously maintained and adapted to a changing operational environment.



Eight Laws of Software Evolution

- **6. Feedback System**– The software systems constitute multi-loop, multi level feedback system and must be treated as such to be successfully modified or improved.



Eight Laws of Software Evolution

- **7. Self Regulation**— A system evolution processes are self – regulating with the distribution of product and process measures close to normal.



Eight Laws of Software Evolution

- **8. Organization Stability**– the average effective global activity rate in an evolving system is invariant over the lifetime of the product.



Case Analysis

- The school's biometric attendance system has been working for five years. Recently, it often fails to detect fingerprints, crashes during peak hours, and is incompatible with new hardware purchased by the school. Teachers say the interface still uses old schedules and cannot accommodate updated class times.
- Which **Laws of Software Evolution** can be observed in this scenario?
 - (Ex: Continuing Change, Increasing Complexity, Reducing Quality)
- Explain how these laws apply.
- What recommendations can help improve or evolve the system?

Case Analysis

A bank released a new update for its mobile banking app. After the update, users complain that the app loads slower, the new design is confusing, and important buttons are harder to find. Some loyal users want the old interface back. Developers report that they added many new features at once, causing unexpected bugs.

- **Guide Questions:**
- Which **Laws of Software Evolution** are reflected in this situation?
 - (Ex: Conservation of Familiarity, Increasing Complexity, Feedback System)
- Describe how the laws are manifested.
- What should developers do to address user feedback and restore app stability?

Case Analysis

A small business developed an inventory system when they only had one store. After two years, the business expanded to three branches. The current software cannot consolidate sales from all branches, lacks multi-user access, and the database becomes slow when handling larger records. The owners want the system enhanced rather than replaced.

- **Guide Questions:**
- Identify the **Laws of Software Evolution** shown here.
 - (Ex: Continuing Growth, Self-Regulation, Organization Stability)
- Explain how these laws apply to the expansion.
- What improvements should be made to ensure the software evolves with the business?